PREY-PREDATOR IN A MAZE WITH REINFORCEMENT LEARNING

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INTRODUCCION

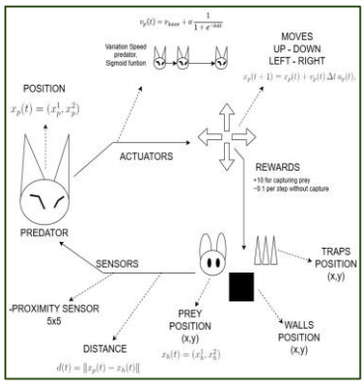
Prey-Predator in a maze is a complex system that involves two autonomous cybernetic agents that interact in a environment like a maze.
Each agents has been trained with reinforcement learning to maximize its objective.
Prey tries maximize the survival time.
Predator tries minimize hunting time.

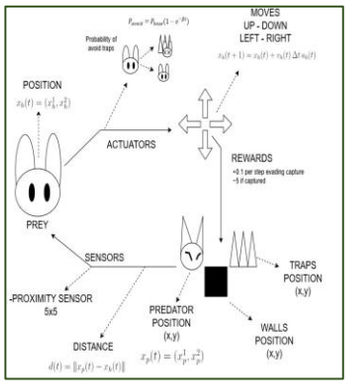
GOAL

We design a system using AI principles and cybernetic foundations. Our goal is develop a functional system with two cybernetics agents in a competition scenery.

METHODOLOGY

Starting with the design of one adaptative agent.
-Receive inputs by virtual sensors.
-Process information in subsystems.
-Take a decision.
-Acts with actuators that affect the env.





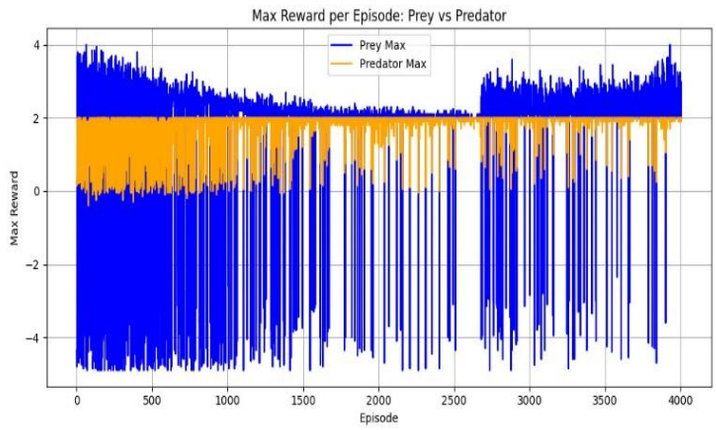
Implementing rewards and penalties for guide the behavior from agents.

RESULTS

Both agents had convergent behaviors, but not expected values behaviors.

-Prey learnt how get great rewards moving in constantly but not in moving over traps.

-Predator learnt to maintain the a constant reward without needed to hunt.



CONCLUSION

The project is a functional system for study this kind of systems. Have two functional cybernetic agents that learn with the perception of the environment. But need improves for get the expected behavior in a competitive MARL.

REFERENCES

[1] P. Phillips, "Reinforcement Learning In Two Player Zero Sum Simultaneous Action Games," arXiv preprint arXiv:2110.04835, 10-Oct-2021. [Online]. Available: https://arxiv.org/abs/2110.04835. [Accessed: 11 Jul 2025].

[2] M. Noguer I Alonso y A. Mfougouon Njupoun, "Game Theory and Multi-Agent Reinforcement Learning: A Mathematical Overview," SSRN, 14 Aug 2024. [Online]. Available: https://ssrn.com/abstract=4926191. [Accessed: 11 Jul 2025].